CLASS 10 — SCIENCE · FORMULA SHEET

Chapter 7: **How do Organisms Reproduce?** | Quick revision, one page

Asexual reproduction

- Fission: binary (Amoeba, Leishmania has a definite plane) and multiple (Plasmodium). Fragmentation: Spirogyra.
- Regeneration: Hydra, Planaria (specialised cells form new tissue). Budding: Hydra, yeast. Spore formation: Rhizopus.
- Vegetative propagation: cutting, layering, grafting, tissue culture; Bryophyllum (leaf), potato (eyes), rose, banana, jasmine.
- Single parent, no gametes, offspring identical; small variations arise from errors in DNA copying.

Sexual reproduction in flowering plants

- Flower: sepals, petals, stamen (anther + filament; male), pistil/carpel (stigma, style, ovary; female).
- Pollination (anther $\rightarrow$ stigma): self or cross. Pollen tube grows through the style to the ovule.
- Fertilisation: male gamete + egg $\rightarrow$ zygote $\rightarrow$ embryo (in ovule). After fertilisation: ovule $\rightarrow$ seed, ovary $\rightarrow$ fruit, other parts wither.
- Seed: embryo (plumule, radicle), cotyledon(s), seed coat. Germination gives a seedling.

Human reproduction

- Male: testes (sperms and testosterone), scrotum (lower temperature), vas deferens, prostate and seminal vesicles (secretions), urethra.
- Female: ovary (eggs and hormones), oviduct (fertilisation site), uterus (implantation), cervix, vagina.
- Zygote $\rightarrow$ embryo $\rightarrow$ foetus; placenta: provides oxygen and nutrients and removes wastes; gestation about 9 months.
- Puberty: boys, facial hair and voice change; girls, breast development and menstruation (about monthly if the egg is not fertilised).
- Number of organisms after n binary fissions $= 2^n$.

MASTERMIND